

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – PSEUDOCODE WRITING TASK

Course Code:	CSA0305	Course Title:	Data Structures
CO Assessed:	CO5 – Naturalization (BL4, BL6)	Assessment:	Assessment Tool 1 – Pseudocode Writing Task
Weightage:	35% of CIA	Total Marks:	100
CO5 Statement:	Construct MST using shortest path algorithms and traverse the graph to model and analyse real-world problem.		
Student Name:	N.S.V. Trivikram	Reg. No.:	192525063
Section / Batch:	AI and DS / 2025	Date:	31/08/2026

Code of Conduct

I N.S.V. Trivikram(192525063) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: N.S.V.Trivikram

Instructions

- This test contains 5 Pseudocode Writing Task questions. Total: 100 Marks.
- When a student answer using pseudocode writing task should evaluate based on the ability to design clear, logical, and efficient pseudocode solutions for data structure problems.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Minimum Spanning Tree	Kruskal's Algorithm	1	20
Graph Sorting	Topological Sort	2	20
Shortest Path Algorithm	MST & Dijkstra's Algorithm	3	20
Shortest Path Algorithm	Dijkstra's Algorithm	4	20
Minimum Spanning Tree	Prim's or Kruskal's Algorithm	5	20
GRAND TOTAL:			100 Marks

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and clear understanding; handles edge cases effectively	Good understanding; minor gaps in edge case handling	Partial understanding; misses some requirements	Misinterprets problem; major gaps
Code Correctness	30	Fully correct; passes all test cases	Mostly correct; minor errors	Partially correct; several test cases fail	Incorrect or non-functional code
Code Quality & Readability	20	Clean, modular, well-commented, follows standards	Readable with some structure	Limited readability; poor structure	Unorganized and hard to read
Complexity Analysis	20	Accurate time & space complexity with justification	Correct but limited explanation	Incomplete or partially correct	Missing or incorrect
Testing & Validation	10	Thorough testing including edge cases	Adequate testing with few edge cases	Minimal testing	No testing evidence

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

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QUESTIONS

Q1
Minimum Spanning Tree -
Kruskal's Algorithm
20 marks

Q2
Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q2 Graph - Topological Sort20 marks

A DAG represents task dependencies. Write pseudocode for Topological Sort using DFS with cycle detection.
Constraints:
a. Detect cycle before sorting
b. Use DFS approach

Your Answer
TOPOLOGICAL_SORT(Graph G):
 for each vertex v in G:
 color[v] = WHITE
 stack = empty stack

 for each vertex v in G:

Save Answer

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Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q1 Minimum Spanning Tree - Kruskal's Algorithm20 marks

Write pseudocode for constructing an MST using Kruskal's Algorithm. A city planning department needs to connect multiple zones with underground fiber-optic cables. Each zone is represented as a vertex, and possible cable connections between zones are represented as edges with installation costs (weights).
Constraints:
a. All zones must be connected
b. Total installation cost must be minimized
c. No redundant connections (no cycles) allowed
d. The network must remain fully connected

Your Answer
parent[rootU] = rootV
ELSE IF rank[rootU] > rank[rootV]:
 parent[rootV] = rootU
ELSE:
 parent[rootV] = rootU
 rank[rootU] = rank[rootU] + 1

Save Answer

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Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q3 Shortest Path Algorithm - Minimum Spanning Tree & Dijkstra's20 marks

Write pseudocode integrating MST + Dijkstra for Smart transportation system.
Constraints:
a. Use MST for infrastructure
b. Use shortest path for routing

Your Answer

```
ShortestPath = FindShortestRoute(MST, source, destination)

DISPLAY "Optimized Infrastructure:", MST
DISPLAY "Shortest Route:", ShortestPath

END
```

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Dijkstra's Algorithm
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Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q4 Shortest Path Algorithm - Dijkstra's Algorithm20 marks

Shortest Path with "Toll-Free" Voucher: You have a voucher that allows you to traverse one edge for free (weight = 0). Write pseudocode to find the shortest path from S to D utilizing this voucher optimally.

Your Answer

```
if used == 0:
    if dist[u][0] + 0 < dist[v][1]:
        dist[v][1] = dist[u][0]
        pq.push((dist[v][1], v, 1))

return MIN(dist[D][0], dist[D][1])
```

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Graph - Topological Sort
20 marks

Q3
Shortest Path Algorithm -
Minimum Spanning Tree &
Dijkstra's
20 marks

Q4
Shortest Path Algorithm -
Dijkstra's Algorithm
20 marks

Q5
Minimum Spanning Tree -
Prim's or Kruskal's Algorithm
20 marks

Q5 Minimum Spanning Tree - Prim's or Kruskal's Algorithm20 marks

Second Best Spanning Tree: Design pseudocode to find the "Second Best" MST.
Hint: Find the MST first, then for every edge in the MST, try removing it and finding the next best edge to replace it.

Your Answer

4. For each edge 'e' in the MST:
a. Create a temporary graph by removing edge 'e'.
b. Find the MST of this new graph.
c. Calculate the total weight of this new MST.
d. If this weight is greater than the original MST weight, update 'second_best_weight' with the minimum value found so far.
5. Return 'second_best_weight'.

Save Answer